भारतीय मानक
Indian Standard

IS 1180 (Part 1) : 2014
[Amalgating IS 1180 (Part 2) : 1989]
(Reaffirmed 2019)

बाहय-रंग तेल इम्मेर्सड वितरण ट्रांसफार्मर तक 2 500 kVA, 33 kV — विशिष्टि

भाग 1 मिनरल तेल-निमिज्जित

(चौथा पुनरीक्षण)

Outdoor Type Oil Immersed Distribution Transformers Upto and Including 2 500 kVA, 33kV — Specification

Part 1 Mineral Oil Immersed

(Fourth Revision)

ICS 29.180

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Transformers Sectional Committee, ET 16

FOREWORD

This Indian Standard (Fourth Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Transformers Sectional Committee, had been approved by the Electrotechnical Division Council.

The Scope of IS 1180 (Part 1) and IS 1180 (Part 2) was to address distribution Transformers primarily of REC (Rural Electrification Corporation Ltd) range, up to 100 kVA, 11 kV. For other range of distribution transformers, reference was made to IS 2026 'Power transformers'.

Distribution Transformer Industry in the country has progressed a lot over the years. In recent times, conservation of energy has assumed importance and minimum energy performance standards for distribution transformers right up to 2 500 kVA 3 phase ratings as well as single phase high voltage distribution transformers up to 25 kVA has been specified in certain regulation. A need was felt to align the standards IS 1180 (Part 1) and IS 1180 (Part 2) in terms of maximum losses with the optimum minimum energy performance standard and also update the standard to take into account of the experience gained since the last revision.

IS 1180 (Part 1) 'Outdoor type three-phase distribution transformers upto and including 100 kVA 11 kV: Part 1 Non-sealed type (*third revision*)' was originally published in 1958, first revised in 1964, second revision in 1981 and subsequently third revision in 1989, whereas IS 1180 (Part 2) 'Outdoor type three-phase distribution transformers upto and including 100 kVA 11 kV: Part 2 Sealed type (*first revision*)', was originally published in 1979, first revision in 1989. Revision of these standards has been undertaken to take into account many technological developments taken place on transformer and also to take into account the developments at international level.

During this revision scope of both standards Part 1 and Part 2 have now been clubbed to make one combined standard for distribution transformer and designated as IS 1180 (Part 1). With the publication of this standard, IS 1180 (Part 2) would be withdrawn. In this revision maximum losses at 50 and 100 percent loading have been incorporated and the scope is extended up to 2 500 kVA. Further, single phase high voltage (11 to 33 kV) distribution transformers up to 25 kVA rating, have also been included to make it a comprehensive standard on Distribution Transformers.

Although IS 2026 'Power Transformers' includes the range of transformers covered by this standard, yet, it has been decided to publish this standard, considering the advantages of having a standard which has simplified the requirements in the distribution transformer range covered by it. However, for various common requirements, normative references have been made to IS 2026 which is, therefore, a necessary adjunct to this standard.

In the preparation of this standard, assistance has been derived from REC Specifications on distribution transformers, R-APDRP Technical Specification for such transformers, BEE recommendations for loss levels of 3 star, 4 star and 5 star rated distribution transformers, Guidelines for Specifications of Energy Efficient Outdoor Type Three Phase and Single Phase Distribution Transformers published by CEA and CBIP Manual on Transformers, Publication 317.

This standard recommends multiple rating with regard to Energy Efficiency that is, three energy efficiency levels: level 1, level 2 and level 3 of transformers corresponding to 3 star, 4 star and 5 star labelled transformers respectively, as prescribed by BEE. It is expected that transformers shall conform to at least energy efficiency level 1. In due course of time with improvements in technology and materials, higher levels of energy efficient transformers shall be progressively used.

This standard (Part 1) is for distribution transformer filled with mineral oil. As mineral oil is semi bio-degradable it is expected that in future from environmental considerations, use of Natural/Synthetic Ester Oils shall increase. This standard also allows use of such oils subject to agreement between user and supplier. When sufficient

(*Continued on third cover*)

Indian Standard

OUTDOOR TYPE OIL IMMERSED DISTRIBUTION TRANSFORMERS UPTO AND INCLUDING 2 500 kVA, 33kV — SPECIFICATION

PART 1 MINERAL OIL IMMERSED

(Fourth Revision)

1 SCOPE

This standard specifies the requirements and tests including standard loss levels of mineral oil-immersed, natural air-cooled, outdoor type, double-wound distribution transformers for use in power distribution systems with nominal system voltages up to and including 33 kV and of following types and ratings:

a) Three phase ratings up to and including 200 kVA both non-sealed type and sealed type.
b) Three phase ratings higher than 200 kVA, up to and including 2 500 kVA both non-sealed type and sealed type.
c) Single phase ratings up to and including 25 kVA sealed type.

2 REFERENCES

The standards listed in Annex A contain provisions which, through reference in this text, constitute provisions of this standard. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards listed in Annex A.

3 TERMINOLOGY

For the purpose of this standard, the following terms and definitions shall apply in addition to those given in IS 1885 (Part 38).

3.1 Distribution Transformer — A distribution transformer is a transformer that provides the final voltage transformation in the electric power distribution system, stepping down the voltage used in the distribution lines to the level used by the customers.

NOTE — The distribution line voltages are 3.3 kV, 6.6 kV, 11 kV, 22 kV and 33 kV in the country. The domestic supply for the consumers is 415 volt, 3 Phase (240 volt, 1 phase), 50 Hz. Transformers with primary voltages of 3.3, 6.6, 11, 22 or 33 kV and secondary voltage of 433 volt, 3 Phase (and 250 volt single phase) are called Distribution Transformers. The maximum rating of these transformers for the purpose of this standard is considered up to 2 500 kVA, 3 Phase.

3.2 Non-Sealed Type Transformer — A transformer which has a conservator and a breather for breathing out and breathing in with expansion and contraction of oil with temperature. The transformer tank body and cover are bolted/clamped/welded type. The tank can also be of corrugated construction.

3.3 Sealed Type Transformer — A transformer which is non-breathing that is so sealed that normally there is no significant interchange between its contents and the external atmosphere. No conservator is provided. Such a transformer may or may not have a cushion of dry air (or inert gas for example; Nitrogen, IS 1747).

Sealed transformers fall in to two categories:

a) Transformers in which the total volume of oil together with air (or inert gas) or any combination thereof, remains constant over the temperature range.
b) Transformers in which the total volume of oil, air (or inert gas) or any combination thereof, varies over the temperature range and this variation is accommodated by a sealed flexible container (corrugated tank) or a flexible membrane.

Sealed type transformers usually have a bolted/clamped / welded cover construction.

4 SERVICE CONDITIONS

The provisions of IS 2026 (Part 1) shall apply.

5 GENERAL

Technical parameters including standard loss levels of three categories of distribution transformers are given in **6**, **7** and **8**.

Other requirements as described in **9** to **22** are applicable for all types and ratings of distribution transformers.

6 TECHNICAL PARAMETERS OF THREE PHASE DISTRIBUTION TRANSFORMERS UP TO AND INCLUDING 200 kVA (NON-SEALED AND SEALED TYPE)

6.1 Ratings

The standard ratings shall be as per Table 1.

Table 1 Standard Ratings

(*Clause* 6.1)

Sl No.	Nominal System Voltage	Standard Ratings (kVA)
(1)	(2)	(3)
i)	Up to and including 11 kV	*6.3, *10,16, *20, 25, *40, 63, 100, 160 and 200
ii)	Above 11 kV up to and including 22 kV	63, 100, 160 and 200
iii)	Above 22 kV up to and including 33 kV	100, 160 and 200

NOTE — *ratings are non-preferred.

6.2 Rated Frequency

The rated frequency shall be 50 Hz.

6.3 Nominal System Voltage

Nominal system voltage shall be chosen from the following:

High Voltage (HV) — 3.3, 6.6, 11, 22 and 33 kV

Low Voltage (LV) — 415V

6.4 Basic Insulation Level (BIL)

6.5 Rated basic insulation level shall be as given in Table 2.

Table 2 Rated Basic Insulation Level

(*Clause* 6.4)

Sl No.	Nominal System Voltage (kV)	Rated BIL (kVp)
(1)	(2)	(3)
i)	3.3	40
ii)	6.6	60
iii)	11	75
iv)	22	125
v)	33	170

6.5 No-Load Voltage Ratios

The no-load voltage ratios shall be as follows:

3 300/433-250, 6 600/433-250, 11 000/433-250, 22 000/433-250 and 33 000/433-250 V

NOTE — Secondary voltage may be selected as 415-240 V, subject to agreement between the user and the supplier

6.6 Winding Connections and Phase Displacement

The primary winding shall be connected in delta and the secondary winding in star [vector symbol, Dyn 11, *see* IS 2026 (Part 1)], so as to produce, a positive phase displacement of 30° from the primary to the secondary vectors of the same phase. The neutral of the secondary winding shall be brought out to a separate insulated terminal.

6.7 Tapping Range and Tapping Methods

6.7.1 No taps are normally required to be provided upto 100 kVA rating, unless specifically specified by the user.

6.7.2 The standard tapping range, when taps are provided above 100 kVA rating shall be as follows:

Winding tapped	HV
Number of tap positions	4
Voltage variation	+2.5 percent to –5 per cent of HV in steps of 2.5 percent

6.7.3 Off circuit tap-changing arrangement shall be either by means of links or by means of an externally-operated switch with mechanical locking device and a position indicator. Arrangement for pad-locking shall be provided.

6.8 Losses and Impedance Values

6.8.1 *Losses —Multiple Rating with Regard to Energy Efficiency*

6.8.1.1 For transformers of HV voltage up to 11 kV, the total losses (no-load + load losses at 75°C) at 50 percent of rated load and total losses at 100 percent of rated load shall not exceed the maximum total loss values given in Table 3.

6.8.1.2 For transformers having voltage class above 11 kV and up to and including 22 kV, the permissible total loss values shall not exceed by 5 percent of the maximum total loss values mentioned in Table 3.

6.8.1.3 For transformers having voltage class above 22 kV and up to and including 33 kV, the permissible total loss values shall not exceed by 7½ percent of the maximum total loss values mentioned in Table 3.

6.8.2 *Impedance*

The recommended impedance at 75°C for different ratings is as per Table 3.

Table 3 Maximum Total Losses Upto 11kV Class Transformers
(*Clauses* 6.8.1.1, 6.8.1.2, 6.8.1.3 *and* 6.8.2)

Sl No.	Rating (kVA)	Impedance (Percent)	Maximum Total Loss (W)					
			Energy Efficiency Level 1		*Energy Efficiency Level 2*		*Energy Efficiency Level 3*	
			50 % Load	100 % Load	50 % Load	100 % Load	50 % Load	100 % Load
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	16	4.5	150	480	135	440	120	400
ii)	25	4.5	210	695	190	635	175	595
iii)	63	4.5	380	1 250	340	1 140	300	1 050
iv)	100	4.5	520	1 800	475	1 650	435	1 500
v)	160	4.5	770	2 200	670	1 950	570	1 700
vi)	200	4.5	890	2 700	780	2 300	670	2 100

NOTE — For non-preferred ratings of Table 1, maximum losses are subject to agreement between the user and the supplier.

6.9 Permissible Flux Density and Over Fluxing

6.9.1 The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency shall not exceed 1.9 Tesla.

NOTE — The design calculations in support of flux density shall be furnished by the manufacturer.

6.9.2 No-load current up to 200 kVA shall not exceed 3 percent of full load current and will be measured by energizing the transformer at rated voltage and frequency. Increase of 12.5 percent of rated voltage shall not increase the no load current by 6 percent maximum of full load current.

6.10 Limits of Temperature-rise

6.10.1 The type of cooling shall be type ONAN as per IS 2026 (Part 2).

6.10.2 The permissible temperature-rise shall not exceed the limits of 40°C (when measured by resistance method) for transformer winding and 35°C (measured by thermometer) for top oil when tested in accordance with IS 2026 (Part 2).

7 TECHNICAL PARAMETERS OF THREE PHASE DISTRIBUTION TRANSFORMERS HIGHER THAN 200 kVA UP TO AND INCLUDING 2 500 kVA (NON-SEALED AND SEALED TYPE)

7.1 Ratings

The standard ratings shall be as per Table 4.

7.2 Rated Frequency

The rated frequency shall be 50 Hz.

7.3 Nominal System Voltage

Nominal system voltage shall be chosen from the following:

HV — 3.3, 6.6, 11, 22 and 33 kV

LV — 415V

Table 4 Standard Ratings
(*Clause* 7.1)

Sl No.	Nominal System Voltage	Standard Ratings (kVA)
(1)	(2)	(3)
i)	Up to and including 11 kV	250, 315, 400, 500, 630, 1 000, 1 250, 1 600, 2 000 and 2 500
ii)	Above 11 kV up to and including 22 kV	250, 315, 400, 500, 630, 1 000, 1 250, 1 600, 2 000 and 2 500
iii)	Above 22 kV up to and including 33 kV	250, 315, 400, 500, 630, 1 000, 1 250, 1 600, 2 000 and 2 500

7.4 Rated Basic Insulation Level (BIL)

The rated basic insulation level (BIL) shall be as given in Table 5.

Table 5 Rated Basic Insulation Level
(*Clause* 7.4)

Sl No.	Nominal System Voltage (kV)	Rated Basic Insulation Level (kVp)
(1)	(2)	(3)
i)	3.3	40
ii)	6.6	60
iii)	11	75
iv)	22	125
v)	33	170

7.5 No-Load Voltage Ratios

The no-load voltage ratios shall be as follows:

3 300/433-250, 6 600/433-250, 11 000/433-250, 22 000/433-250 and 33 000/433-250 V

NOTE — Secondary voltage may be selected as 415-240 V, subject to agreement between the user and the supplier.

7.6 Winding Connections and Phase Displacement

The primary winding shall be connected in delta and the secondary winding in star [vector symbol, Dyn 11, *see* IS 2026 (Part 1)], so as to produce, a positive phase displacement of 30° from the primary to the secondary vectors of the same phase. The neutral of the secondary winding shall be brought out to a separate insulated terminal.

Alternatively, [Dyn1, *see* IS 2026 (Part 1)] can also be specified. If system and application requirements demand different vector groups, the same can also be adopted.

7.7 Tapping Ranges and Tapping Methods

7.7.1 The standard tapping ranges, when taps are provided, shall be as follows:

Winding tapped	HV
Number of tap positions	7
Voltage variations	+ 5 percent to –10 percent in steps of 2.5 percent for variation of HV

7.7.2 Off circuit tap-changing arrangement shall be either by means of links or by means of an externally-operated switch with mechanical locking device and a position indicator. Arrangement for pad-locking shall be provided.

7.7.3 For ratings greater than 500 kVA, on load tap changers may be provided for variation of HV voltage from +5 percent to –15 percent in steps of 2.5 percent.

7.8 Losses and Impedance Values

7.8.1 *Losses — Multiple Rating with Regard to Energy Efficiency*

7.8.1.1 For transformers of HV voltage up to 11 kV, the total losses (no-load + load losses at 75°C) at 50 percent of rated load and total losses at 100 percent of rated load shall not exceed the maximum total loss values given in Table 6.

7.8.1.2 For transformers having voltage class above 11 kV and up to and including 22 kV, the permissible total loss values shall not exceed by 5 percent of the maximum total loss values mentioned in Table 6.

7.8.1.3 For transformers having voltage class above 22 kV and up to and including 33kV, the permissible total loss values shall not exceed by 7½ percent of the maximum total loss values mentioned in Table 6.

7.8.2 *Impedance*

The recommended percent impedance at 75°C for different ratings shall be as per Table 6.

7.9 Permissible Flux Density and Overfluxing

7.9.1 The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla.

NOTE — The design calculations in support of flux density shall be furnished by the manufacturer.

7.9.2 No load current shall not exceed 2 percent of the full load current and shall be measured by energizing the transformer at rated voltage and frequency. Increase of 12.5 percent of rated voltage shall not increase the no-load current by 5 percent of full load current.

7.10 Limits of Temperature-rise

7.10.1 The type of cooling shall be ONAN as per IS 2026 (Part 2).

7.10.2 The permissible temperature-rise shall not exceed the limits of 45°C when measured by resistance method for transformer winding and 40°C measured

Table 6 Maximum Total Losses Up to 11 kV Class Transformer

(*Clause* 7.8.1.1)

Sl No.	Rating (kVA)	Impedance (Percent)	Maximum Total Loss (W)					
			Energy Efficiency Level 1		*Energy Efficiency Level 2*		*Energy Efficiency Level 3*	
			50% Load	100% Load	50% Load	100% Load	50% Load	100% Load
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	250	4.50	1 050	3 150	980	2 930	920	2 700
ii)	315	4.50	1 100	3 275	1 025	3 100	955	2 750
iii)	400	4.50	1 300	3 875	1 225	3 450	1 150	3 330
iv)	500	4.50	1 600	4 750	1 510	4 300	1 430	4 100
v)	630	4.50	2 000	5 855	1 860	5 300	1 745	4 850
vi)	1 000	5.00	3 000	9 000	2 790	7 700	2 620	7 000
vii)	1 250	5.00	3 600	10 750	3 300	9 200	3 220	8 400
viii)	1 600	6.25	4 500	13 500	4 200	11 800	3 970	11 300
ix)	2 000	6.25	5 400	17 000	5 050	15 000	4 790	14 100
x)	2 500	6.25	6 500	20 000	6 150	18 500	5 900	17 500

by thermometer for top oil when tested in accordance with IS 2026 (Part 2).

8 TECHNICAL PARAMETERS OF SINGLE PHASE DISTRIBUTION TRANSFORMERS UP TO AND INCLUDING 25 kVA (SEALED TYPE)

8.1 Ratings

The standard ratings shall be as per Table 7.

Table 7 Standard Ratings
(*Clause* 8.1)

Nominal System Voltage	Standard Ratings (kVA)
(1)	(2)
11 kV	5,10,16 & 25
22 kV	10,16 & 25
33 kV	16 & 25

8.2 Rated Frequency

The rated frequency shall be 50 Hz.

8.3 Nominal System Voltage

Nominal system voltage shall be chosen from the following:

HV — 11, 22 and 33 kV

LV — 415V (240 V, 1 Phase)

8.4 Rated Basic Insulation Level (BIL)

Rated basic insulation level shall be as given in Table 8.

Table 8 Rated Basic Insulation Level
(*Clause* 8.4)

Nominal System Voltage (kV)	Rated BIL (kVp)
(1)	(2)
11	75
22	125
33	170

8.5 No-Load Voltage Ratio

The no-load voltage ratios shall be as follows:

$11\,000/\sqrt{3}$ / 250 V , 11 000 / 250 V

$22\,000/\sqrt{3}$ / 250 V , 22 000 / 250 V

$33\,000/\sqrt{3}$ / 250 V , 33 000 / 250 V

NOTE — Secondary voltage may be selected as 415-240 V, subject to agreement between the user and the supplier.

8.6 Number of Phases and Polarity

Number of phases shall be one (Single Phase).

Polarity: Additive or Subtractive

8.7 Tap Changing Arrangement

Taps are not required.

8.8 Losses and Impedance Values

8.8.1 *Losses — Multiple Rating with Regard to Energy Efficiency*

8.8.1.1 For transformer of HV voltage 11 kV, the total losses (no load + load losses at 75°C) at the 50 percent of rated load and total losses at 100 percent of rated load shall not exceed the maximum total loss values given in Table 9.

8.8.1.2 For transformers having voltage class above 11 kV and up to and including 22 kV, the permissible total loss values shall not exceed by 7½ percent of the maximum total loss values given in Table 9.

8.8.1.3 For transformers having voltage class above 22 kV and up to and including 33 kV, the permissible total loss values shall not exceed by 10 percent of the maximum total loss values given in Table 9.

8.8.2 *Impedance*

The recommended percent impedance at 75°C for different ratings shall be as per Table 9.

8.9 Permissible Flux Density and Overfluxing

8.9.1 The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla.

NOTE — The design calculations in support of flux density shall be furnished by the manufacturer.

Table 9 Maximum Total Losses of Single Phase Transformers Upto 11 kV
(*Clauses* 8.8.1.1, 8.8.1.2 and 8.8.1.3)

Sl No.	Rating (kVA)	Impedance (Percent)	Maximum Total Loss (W)					
			Energy Efficiency Level 1		*Energy Efficiency Level 2*		*Energy Efficiency Level 3*	
			50% Load	100% Load	50% Load	100% Load	50% Load	100% Load
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	5	2.5	40	115	35	95	30	75
ii)	10	4.00	70	190	60	170	55	150
iii)	16	4.00	95	265	82	224	63	190
iv)	25	4.00	125	340	110	300	95	260

8.9.2 No load current shall not exceed 3 percent of full load current and will be measured by energizing the transformer at rated voltage and frequency. Increase of 12.5 percent of rated voltage shall not increase the no-load current by 6 percent maximum of full load current.

8.10 Limits of Temperature-rise

8.10.1 The type of cooling shall be ONAN as per IS 2026 (Part 2).

8.10.2 The permissible temperature-rise shall not exceed the limits of 40°C when measured by resistance method for transformer winding and 35°C measured by thermometer for top oil when tested in accordance with IS 2026 (Part 2).

9 STANDARD MATERIALS

9.1 Major material used in the transformer shall conform to the following Indian Standards:

a) Cold rolled grain oriented electrical steel – IS 3024

b) Amorphous core material – (IS under preparation)

c) Copper/Aluminum conductor – IS 191, IS 1897, IS 7404, IS 12444, IS 13730/IS 6162 series as given in Annex A.

d) Kraft paper – IS 9335 series as given in Annex A.

e) Press board – IS 1576

f) Mineral oil – IS 335

NOTE — Use of other insulating liquids namely natural ester, synthetic organic ester IS 16081 subject to agreement between the user and the supplier.

10 TERMINAL ARRANGEMENT

10.1 For Three Phase Transformers

10.1.1 The transformers shall be fitted on high voltage and low voltage sides with outdoor type bushings of appropriate voltage and current ratings. The high voltage bushings (3 Nos.) shall conform to IS 2099. The low voltage bushings (4 Nos.) shall conform to IS 7421. Alternatively, the low voltage side may be made suitable for adoption of PVC/XLPE cables of suitable size.

10.1.2 If required by the user, a suitable cable-end box may be provided on the high voltage end or low voltage side.

10.1.3 In case of sealed type transformer, the terminal arrangements shall be such that it shall be possible to replace the bushings (external) without opening the cover and also without affecting the sealing of the transformer. The arrangement shall meet the following requirements:

HV and LV Bushings:

The bushing shall be made in two parts. The outer bushing shall be of porcelain. The dimensions of the outer bushing shall conform to relevant Part/Section of IS 3347 depending on the voltage class. The internal bushing shall be of either porcelain or tough insulating material, like epoxy and shall have embedded stem. Metal portion of the internal HV and LV bushing inside the tank shall remain dipped in oil in all operating conditions.

NOTE — Any other suitable arrangement can be used subject to agreement between the user and the supplier.

10.1.4 Gaskets shall be made of synthetic rubber or synthetic rubberized cork resistant to hot transformer oil.

10.1.5 The dimensions of bushings of the following voltage classes shall conform to the following Indian Standards mentioned against them:

Voltage Class	*For Porcelain Parts*	*For Metal Parts*
Up to 1.0 kV bushings	IS 3347 (Part 1/Sec 1)	IS 3347 (Part 1/Sec2)
3.6 kV bushings	IS 3347 (Part 2/Sec 1)	IS 3347 (Part 2/Sec 2)
12 kV bushings	IS 3347 (Part 3/Sec 1)	IS 3347 (Part 3/Sec 2)
24 kV bushings	IS 3347 (Part 4/Sec 1)	IS 3347 (Part 4/sec 2)
36 kV bushings	IS 3347 (Part 5/Sec 1)	IS 3347 (Part 5/Sec 2)

10.2 For Single Phase Transformers

For $11/\sqrt{3}$, $22/\sqrt{3}$, and $33/\sqrt{3}$ transformers, neutral end of the HV winding shall be brought out to Neutral through 1.1 kV bushing. Neutral terminal shall be connected to tank by a tinned copper strip of adequate size.

For 11, 22, 33 kV transformers, two HV bushings shall be used for termination of both ends of HV winding.

The HV bushings shall be fixed to the top cover and the LV bushings of 1.1 kV class shall be fixed to the transformer tank on sides.

10.3 Marking and Relative Positions of Terminals

Appropriate characters in accordance with IS 2026 (Part 1) shall be indelibly marked upon or adjacent to terminals.

11 MINIMUM CLEARANCES IN AIR

The minimum phase-to-phase and phase-to-earth external clearances for LV and HV bushings shall be as per Table 10.

Table 10 External (Air) Clearances Between Bushings Mounted on Transformers
(*Clause* 11)

Nominal System Voltage	Phase-to-Phase Clearance in mm	Phase-to-Earth Clearance in mm
(1)	(2)	(3)
Up to 1.1 kV	75	40
11 kV	255	140
22 kV	330	230
33 kV	350	320

11.1 For transformers with air filled cable-end box/ connection chamber, the phase-to-phase and phase-to-earth clearance shall be as per Table 11.

Table 11 Air Clearances in Cable Box
(*Clause* 11.1)

Nominal System Voltage	Phase-to-Phase Clearance in mm	Phase-to-Earth Clearance in mm
(1)	(2)	(3)
Up to 1.1 kV	25	20
11 kV	130	80
22 kV	240	140
33 kV	350	220

12 CONNECTORS (APPLICABLE FOR BARE BUSHING TERMINATIONS ONLY)

Wherever specified, suitable bimetallic connectors (clamp type) shall be provided on both HV and LV side in order to ensure sound and robust connection.

13 MARKING

13.1 Rating Plate

Each transformer shall be provided with rating plate made of anodized aluminium/stainless steel material securely fixed on the outer body, easily accessible, showing the information given in Fig. 1 for 3 phase transformers and Fig. 2 for single phase transformers. The entries on the rating plate shall be indelibly marked for example, by etching, engraving or stamping.

13.2 Terminal Marking Plate

Each transformer shall be provided with a terminal marking plate in accordance with Fig. 3 to Fig. 5 whichever is applicable.

13.3 The rating and terminal marking plates may be combined into one plate at the option of the manufacturer.

13.4 The distribution transformer may also be marked with the Standard Mark.

13.4.1 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

14 MOUNTING ARRANGEMENT

14.1 The under-base of all three phase transformers upto 200 kVA ratings shall be provided with two 75 mm × 40 mm channels 460 mm long as shown in Fig. 6 to make them suitable for fixing to a platform or plinth.

14.2 The under-base of all transformers beyond 200 kVA shall be as per Fig. 7 to make them suitable for mounting on rollers.

14.3 Suitable pole mounting arrangement may be alternatively provided for 3 phase transformers upto 500 kVA, subject to agreement between user and supplier.

14.4 Single phase transformers are pole mounted type and shall be provided with two mounting lugs suitable for fixing the transformer to a single pole by means of two bolts of 20 mm diameter.

Both mounting lugs are made with steel of minimum 5 mm thickness.

15 TRANSFORMER TANK

15.1 Construction

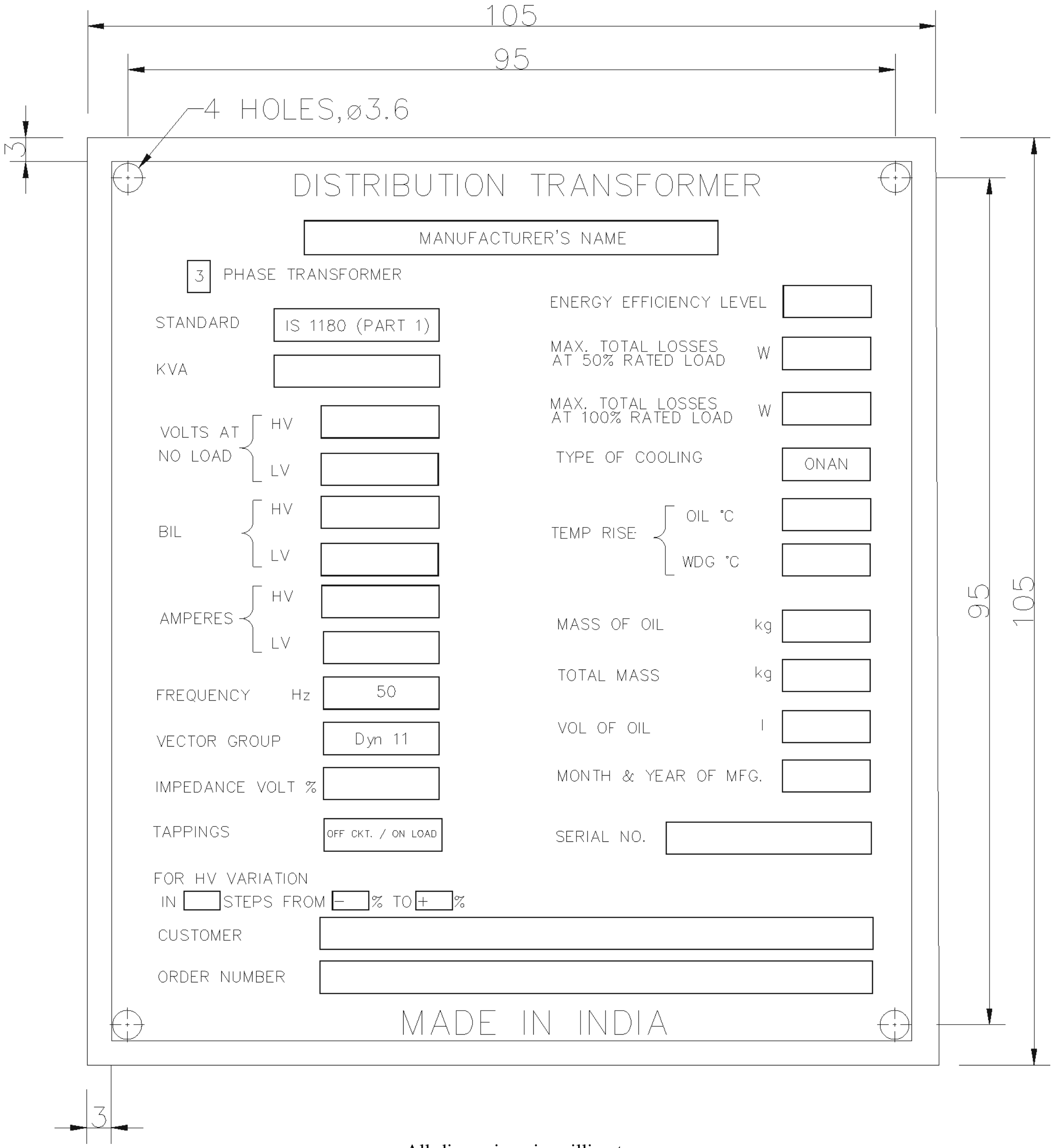
15.1.1 For non-sealed or sealed type transformer, transformer tank can be of plain tank configuration with/without radiator fins or cooling tubes. The tank can also be made of corrugated panels of adequate thickness, also used for cooling. The transformer tank covers shall be bolted/clamped alternatively welded with tank rim so as to make a leak proof joint. The curb design in case of welded construction shall be such that it is possible to remove the weld and reweld the tank at least two times.

15.1.2 The transformer tank shall be of adequate mechanical strength to withstand positive and negative pressures built up inside the tank while the transformer is in operation.

15.1.3 All welding operations shall be carried out by qualified welders.

15.1.4 The tank design shall be such that the core and windings can be lifted freely.

15.1.5 For single phase sealed type transformers, the circular base plate edges of the tank shall be folded



All dimensions in millimetres.

FIG. 1 RATING PLATE FOR THREE PHASE TRANSFORMERS

upward for at least 25 mm, to have sufficient overlap with vertical sidewall of the transformer.

15.2 Pressure and Vacuum Requirements

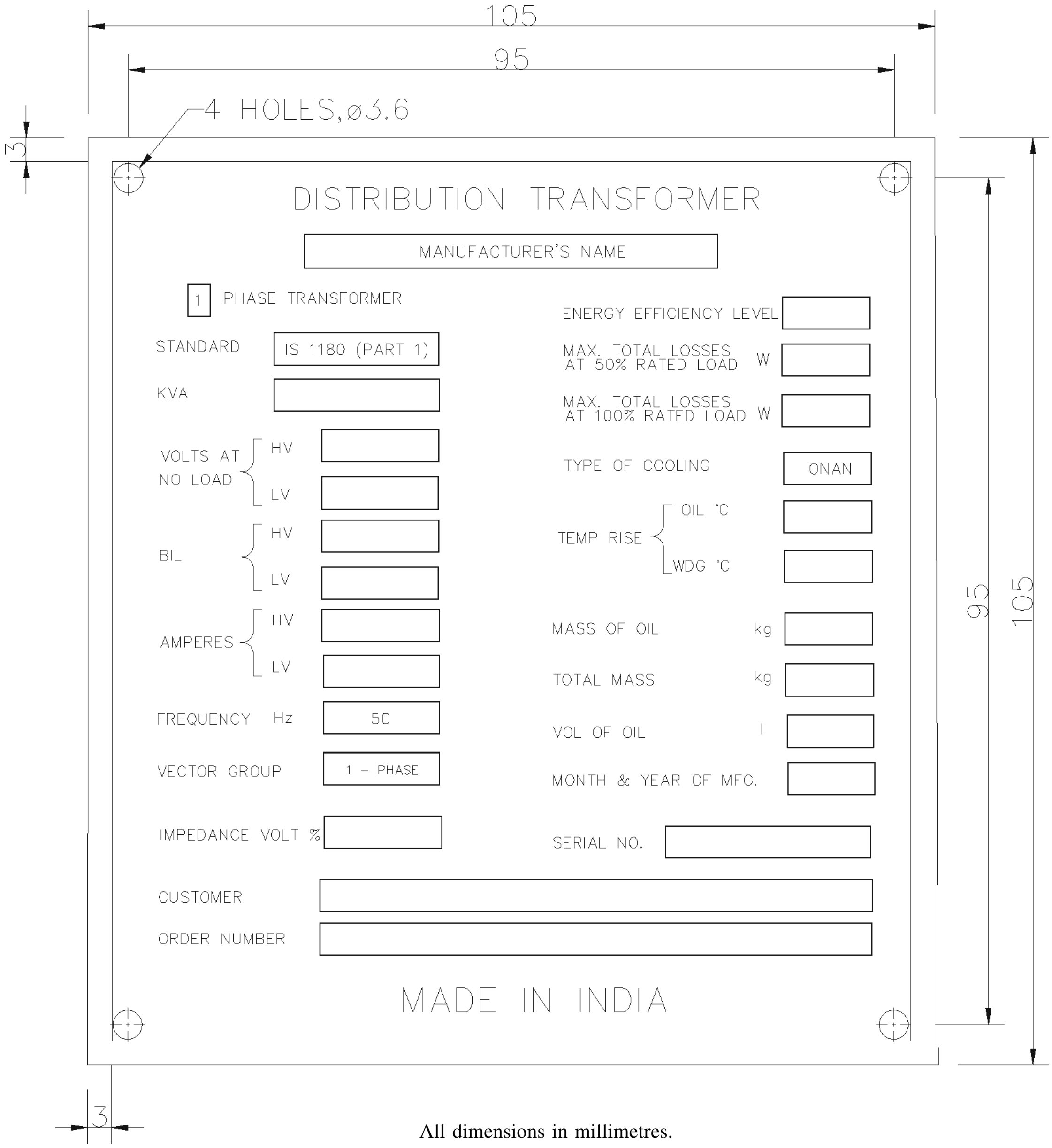
15.2.1 In case of transformers up to 200 kVA, plain tank shall be capable of withstanding a pressure of 80 kPa and a vacuum of 250 mm of mercury. Limiting values of the deflections are specified in **21.5.1**.

For transformers above 200 kVA plain tank shall be capable of withstanding a pressure of 80 kPa and a vacuum of 500 mm of mercury. Limiting values of the deflections are specified in **21.5.2**.

For single phase transformers up to 25 kVA, the transformer tank shall be of robust construction round in shape and shall be capable of withstanding a pressure of 100 kPa and a vacuum of 760 mm of mercury.

15.2.2 For three phase transformers up to 2 500 kVA, transformer tanks with corrugations shall be designed for a pressure of 15 kPa measured at the top of the tank with no leakage.

15.2.3 For three phase sealed type transformers with cover welded to the curb of the tank shall be of sound and robust construction so as to withstand pressure of 80 kPa without any deformation.



All dimensions in millimetres.

FIG. 2 RATING PLATE FOR SINGLE PHASE TRANSFORMERS

15.2.4 For single phase transformers of sealed type construction, when the space on the top oil is filled with dry air or inert gas, the inert gas plus oil volume inside the tank shall be such that even under extreme operating conditions, the pressure generated inside the tank does not exceed 0.4 kg/cm^2 positive or negative.

15.3 All bolts/nuts/washers exposed to atmosphere shall be as follows:

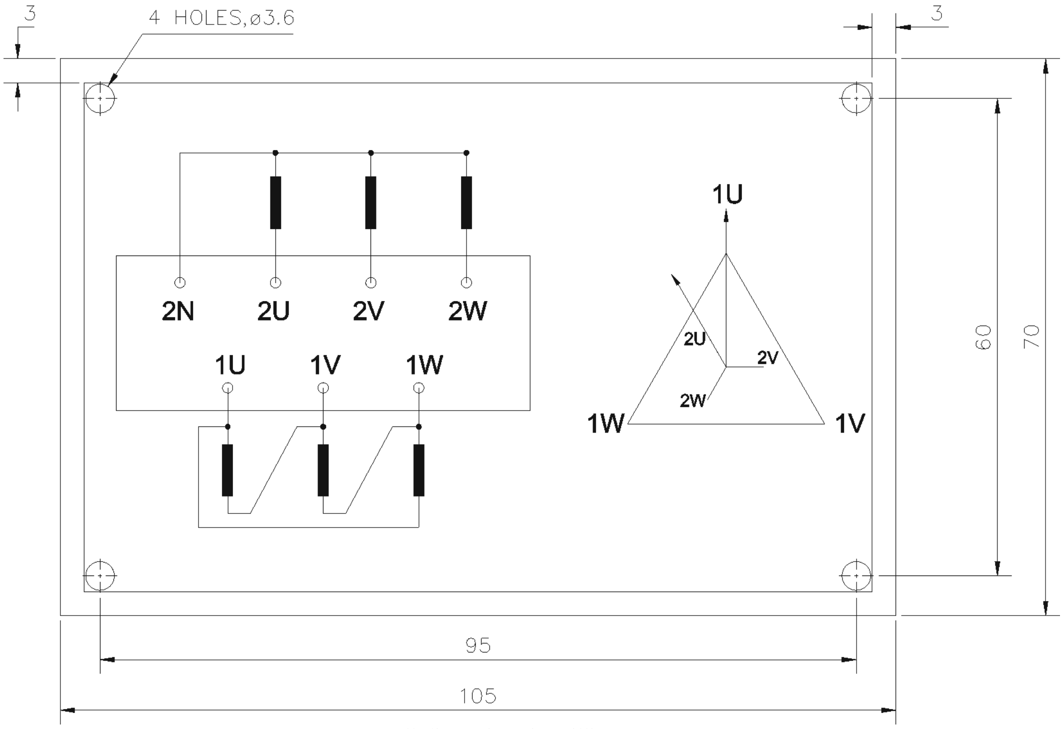
a) Size 12 mm or below — stainless steel.
b) Above 12 mm — steel with suitable finish like electro galvanized with passivation or hot dip galvanized.

15.4 Gaskets wherever used shall conform to Type III as per IS 11149/Type C as per IS 4253 (Part 2).

15.5 Inside of tank shall be painted with varnish or oil resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. Table 12 shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be subject to agreement between the user and the transformer manufacturer.

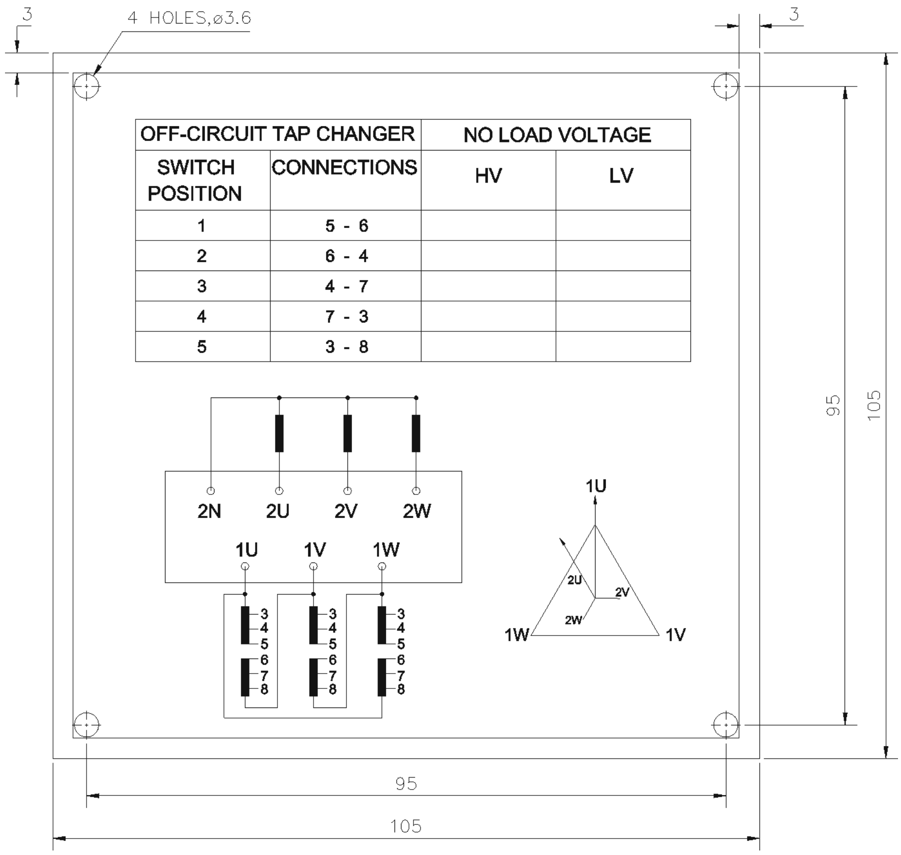
16 CONSERVATOR FOR NON-SEALED TYPE TRANSFORMERS

16.1 Transformers of ratings 63 kVA and above with plain tank construction, the provision of conservator



All dimensions in millimetres.

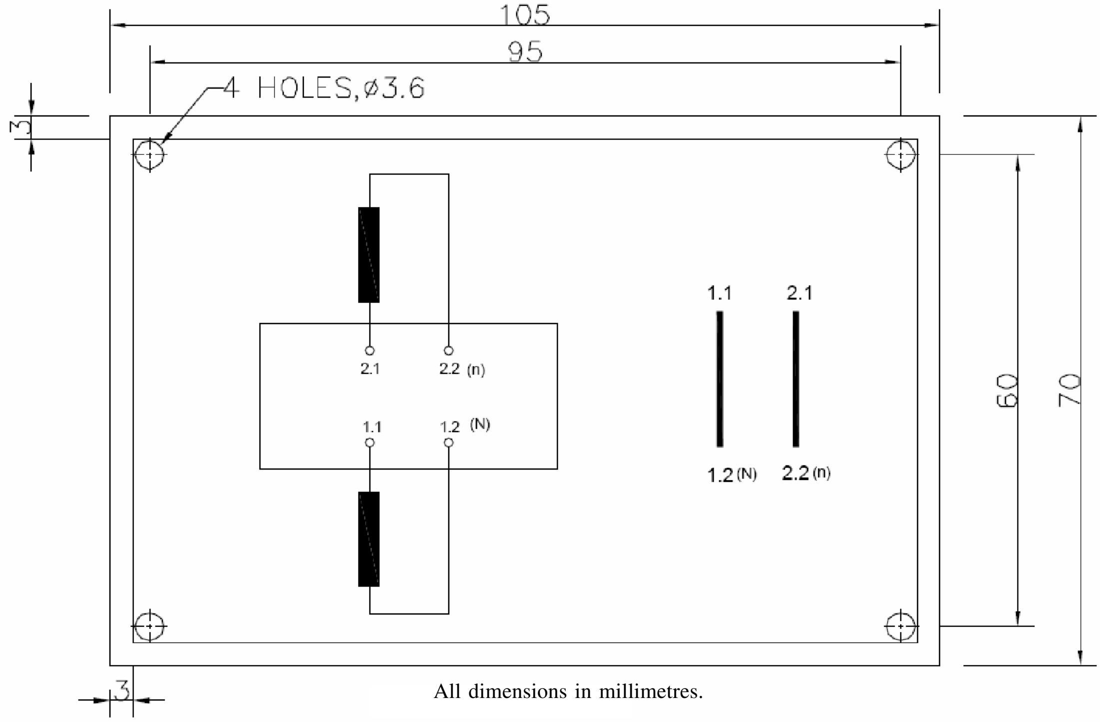
FIG. 3 TERMINAL MARKING PLATE FOR THREE PHASE TRANSFORMERS WITHOUT TAPS



OFF-CIRCUIT TAP CHANGER		NO LOAD VOLTAGE	
SWITCH POSITION	CONNECTIONS	HV	LV
1	5 - 6		
2	6 - 4		
3	4 - 7		
4	7 - 3		
5	3 - 8		

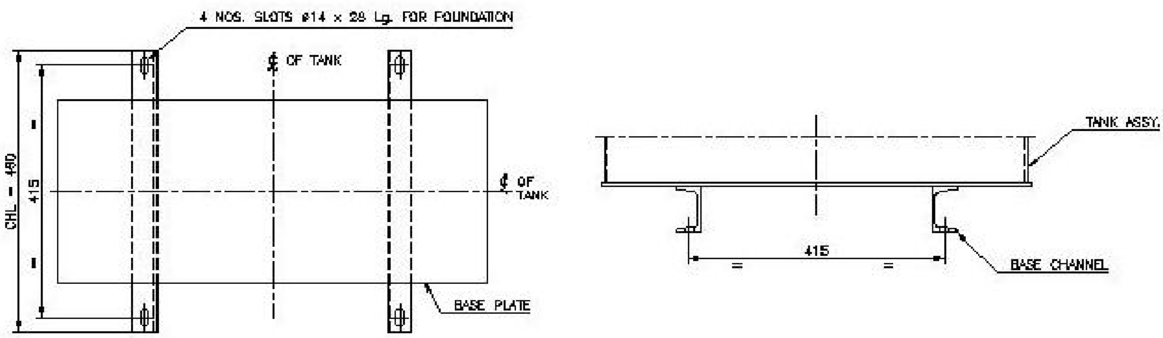
All dimensions in millimetres.

FIG. 4 TERMINAL MARKING PLATE FOR THREE PHASE TRANSFORMERS WITH TAPS



All dimensions in millimetres.

FIG. 5 TERMINAL MARKING PLATE FOR SINGLE PHASE TRANSFORMERS



All dimensions in millimetres.

FIG. 6 MOUNTING DIMENSIONS OF TRANSFORMERS UP TO 200 kVA

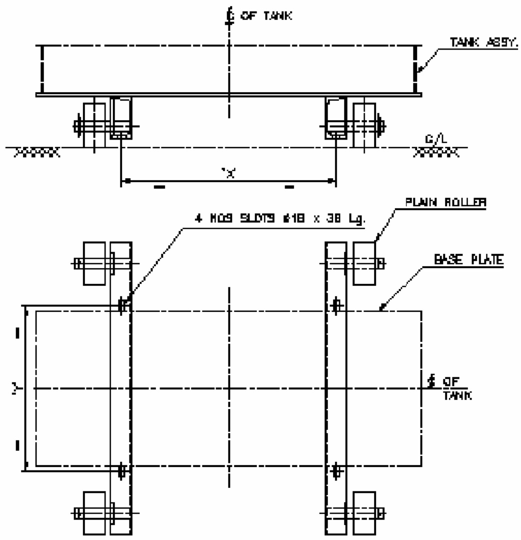
Table 12 Paint Scheme for Distribution Transformers

(*Clause* 15.5)

Sl No.	Paint Type	Area to be Painted	No. of Coats	Total Dry Film Thickness, *Min* (microns)
(1)	(2)	(3)	(4)	(5)
i)	Thermo setting powder paint	Inside	01	30
		Outside	01	60
ii)	Liquid paint			
	a) Epoxy (primer)	Outside	01	30
	b) Polyurethane (finish coat)	Outside	02	25 each
	c) Hot oil resistant paint/varnish	Inside	01	35 / 10

is mandatory. For corrugated tank and sealed type transformers with or without inert gas cushion, conservator is not required.

16.2 When a conservator is provided, oil gauge and the plain or dehydrating breathing device shall be fixed to the conservator which shall also be provided with a



2 500	1 200	800
1 800	1 000	750
1 000	800	750
600	700	600
650	700	800
500	700	450
400	800	450
200	550	415
100	550	415

All dimensions in millimetres.

NOTE — Bidirectional rollers can also be used as per mutual agreement between the user and the supplier.

FIG. 7 MOUNTING DIMENSIONS OF TRANSFORMERS BEYOND 200 kVA

drain plug and a filling hole (1¼″ normal size thread) with cover. The capacity of a conservator tank shall be designed keeping in view the total quantity of oil and its contraction and expansion due to temperature variations. In addition, the cover of main tank shall be provided with an air release plug to enable air trapped within to be released, unless the conservator is so located as to eliminate the possibility of air being trapped within the main tank.

16.3 The inside diameter of the pipe connecting the conservator to the main tank should be 25 to 50 mm and it should be projected into the conservator so that its end is at least 20 mm above the bottom of the conservator so as to create a sump for collection of impurities. The minimum oil level corresponding to –5°C should be above the sump level.

17 ABILITY OF TRANSFORMERS TO WITHSTAND EXTERNAL SHORT CIRCUIT

The performance of transformer under external short-circuit conditions shall be in accordance with IS 2026 (Part 5).

18 EFFICIENCY AND REGULATION

When statements of efficiency and regulations are required they shall be based on specified loading at the rated kVA and unity power factor and computed in accordance with Annex B and Annex C respectively.

NOTE — Efficiency and regulations at other power factors as agreed between the user and supplier shall also be computed.

19 TOLERANCES

The tolerance on electrical performance excluding losses shall be as given in IS 2026 (Part 1).

20 FITTINGS

20.1 Standard Fittings

The following standard fittings shall be provided:

a) Two earthing terminals with the earthing symbol ⏚;

b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature;

NOTES

1 Minimum and maximum positions correspond to the operating temperature of –5°C and 90°C respectively (for non-sealed type transformer).

2 Minimum position corresponds to the operating temperature of 30°C (for sealed type transformers).

c) Air release device (for non-sealed type transformers)

d) Rating and terminal marking plates;

e) Plain breathing device for non-sealed type transformers which would not permit ingress of rain water and insects up to 200 kVA transformers. Above 200 kVA transformers dehydrating breather shall be provided;

f) Drain-*cum*-sampling valve (¾″ nominal size thread, IS 554) preferably steel with plug for three phase transformers;

g) Thermometer pocket with cap;
h) Oil filling holes having (1¼″ nominal size thread) with cover (for sealed type transformers without conservator);
j) An extended pipe connection on upper end with welded cover for sealed type transformers. The pipe should be suitably threaded over a sufficient length to enable use of a refilling/siphon connection after removing the welded cover or any other similar arrangement capable of reuse;

NOTE — The bottom drain valve and filling hole may also be used for filtration purpose.

k) Lifting lugs for the complete transformer as well as for core and winding assembly;
m) Nitrogen/air filling device/pipe with welded cover capable of reuse (for sealed type transformer);
n) Pressure relief device or explosion vent above 200 kVA;
p) One filter valve on the upper side of the tank (for transformers above 200 kVA);
q) Unidirectional flat rollers (for transformers above 200 kVA);
r) Inspection hole (for transformers above 200 kVA);
s) Pressure gauge for sealed transformers with radiators and nitrogen cushion (above 200 kVA);
t) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);
u) LV earthing arrangement for single phase transformers; and
v) Buchholz relay for transformers above 1 000 kVA.

20.2 Optional Fittings

The following shall be available as additional fittings at the option of the user wherever specified:

a) Dehydrating breather in lieu of plain breathing device for transformers up to 200 kVA;
b) Filter valve (1¼″ nominal size thread) for transformers up to 200 kVA;
c) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. for transformers up to 200 kVA;
d) Bird guard;
e) Terminal connectors;
f) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA;
g) Jacking pads (for transformer above 1 600 kVA);
h) Buchholz relay (for transformers above 200 kVA);
j) Magnetic oil level gauge (for transformer above 1 600 kVA) with low oil level alarm contact;
k) Non return valve (for conducting pressure test);
m) Pressure relief device or explosion vent (up to 200 kVA);
n) Protection relay for sealed type transformers for internal parameters that is pressure, temperature, oil level and gas detection (above 1 000 kVA); and
p) 4 Nos anti-theft stainless steel fasteners with breakaway nut shall be provided at top cover (up to 200 kVA).

NOTE — IS 3639 describes some of the fittings and accessories.

21 TESTS

21.1 General

All routine, type and special tests as described in **21.2** to **21.4** shall be performed as per relevant parts of IS 2026. Pressure and oil leakage test shall be conducted as per **21.5**.

21.2 Routine Tests (to be conducted on all units)

The following shall constitute the routine tests:

a) Measurement of winding resistance [IS 2026 (Part 1)].
b) Measurement of voltage ratio and check of phase displacement [IS 2026 (Part 1)].
c) Measurement of short-circuit impedance (principal tapping, when applicable) and load loss at 50 percent and 100 percent load [IS 2026 (Part 1)].
d) Measurement of no-load loss and current [IS 2026 (Part 1)].
e) Measurement of insulation resistance [IS 2026 (Part 1)].
f) Induced over-voltage withstand test [IS 2026 (Part 3)].
g) Separate-source voltage withstand test [IS 2026 (Part 3)].
h) Pressure test (*see* **21.5**).
j) Oil leakage test (*see* **21.5**).

21.3 Type Tests (to be conducted on one unit)

The following shall constitute the type tests:

a) Lightning impulse test [IS 2026 (Part 3)].

b) Temperature-rise test [IS 2026 (Part 2)].

NOTE — Maximum measured total loss (No load at rated excitation + load loss at maximum current tap converted to 75 ºC reference temperature) at 100 percent loading shall be supplied during temperature rise test.

c) Short-circuit withstand test [IS 2026 (Part 5)] (up to 200 kVA).

NOTE — Routine tests before and after short circuit test shall be conducted as per IS 2026 (Part 1).

d) Pressure test (*see* **21.5**)

21.4 Special Tests (to be conducted on one unit)

The following shall constitute the special tests which shall be carried out by mutual agreement between the user and the supplier.

a) Determination of sound levels [IS 2026 (Part 10)]
b) Short-circuit withstand test [IS 2026 (Part 5)] (above 200 kVA)

NOTE — Routine tests before and after short circuit test shall be conducted as per IS 2026 (Part 1).

c) No load current at 112.5 percent voltage (*see* **5.9.3**).
d) *Paint adhesion tests*. The test is performed as per ASTM D3359 (Standard Test Methods for measuring adhesion by Tape test).
e) BDV and moisture content of oil in the transformer (IS 335).

NOTE — Tests at (d) and (e) may be carried out on more than one unit subject to agreement between user and supplier.

21.5 Pressure and Oil Leakage Test

21.5.1 *For Transformers up to 200 kVA*

21.5.1.1 *Pressure test (type test)*

For non-sealed and sealed type transformers, the transformer tank shall be subjected to air pressure of 80 kPa for 30 min and vacuum of 250 mm of mercury for 30 min.

The permanent deflection of flat plates, after pressure/vacuum has been released, shall not exceed the values given below.

Length of Plate	*Deflection*
Up to 750 mm	5 mm
751 to 1 250 mm	6.5 mm

21.5.1.2 *Pressure (routine test)*

a) *Non-sealed type transformers (plain tanks)*

The transformer with bolted cover shall be tested at an air pressure of 35 kPa above atmosphere pressure maintained inside the tank for 10 min. There should be no leakage at any point.

b) *Corrugated tanks*

The corrugated transformer tank shall be tested for air pressure of 15 kPa above atmosphere pressure maintained inside the tank for 10 min. There should be no leakage at any point.

c) *Sealed type transformers*

The transformer with welded cover shall be tested at an air pressure of 80 kPa above atmosphere pressure maintained inside the tank for 10 min. There should be no leakage at any point.

21.5.1.3 *Oil leakage test (routine test)*

The assembled transformer for non-sealed and sealed type with all fittings including bushings in position, shall be tested at a pressure equivalent to twice the normal head measured at the base of the tank for 8 h. There should be no leakage at any point. Tank with corrugations shall be tested for oil leakage test a pressure of 15 kPa measured at the top of the tank for 6 h. There should be no leakage at any point.

21.5.2 *For Transformers Above 200 kVA and Up to Including 2 500 kVA*

21.5.2.1 *Pressure test (Type test)*

For non-sealed and sealed type transformers, the transformer tank subjected to air pressure of 80 kPa for 30 min and vacuum of 500 mm of mercury for 30 min. The permanent deflection of flat plate, after pressure/vacuum has been released, shall not exceed the values given below.

Length of Plate	*Deflection*
Up to 750 mm	5.0 mm
751 mm to 1 250 mm	6.5 mm
1 251 mm to 1 750 mm	8.0 mm
Above 1 751 mm	9.0 mm

21.5.2.2 *Pressure test (routine test)*

a) *Plain tanks*

The transformer tank with welded/bolted cover shall be tested at a pressure of 35 kPa above atmosphere pressure maintained inside the tank for 10 min. There should be no leakage at any point.

b) *Corrugated tanks*

The corrugated transformer tank shall be tested for air pressure of 15 kPa above atmosphere pressure maintained inside the tank for 10 min. There should be no leakage at any point.

21.5.2.3 *Oil leakage test (routine test)*

The assembled transformer for non-sealed and sealed type with all fittings including bushing in position shall be tested at a pressure equivalent to twice the normal head measured at the base of the tank for 8 h. There should be no leakage at any point. Tank with corrugations shall be tested for oil leakage test a pressure of 15 kPa measured at the top of the tank for 6 h. There should be no leakage at any point.

21.5.3 *For Single Phase Distribution Transformers Up to Including 25 kVA*

21.5.3.1 *Pressure test (type test)*

The tank shall be subjected to air pressure of 100 kPa above atmospheric pressure for 30 min. There should be no leakage at any point and there is no deformation of tank.

21.5.3.2 *Pressure (routine test)*

The transformer tank shall be tested at a pressure of 35 kPa above atmosphere pressure maintained inside the tank for 10 min. There should be no leakage at any point.

21.5.3.3 *Oil leakage test (routine test)*

The assembled transformer for non-sealed and sealed type with all fittings including bushings in position, shall be tested at a pressure equivalent to twice the normal head measured at the base of the tank for 6 h. There should be no leakage at any point. Tank with corrugations shall be tested for oil leakage test at a pressure of 15 kPa measured at the top of the tank for 6 h. There should be no leakage at any point.

22 INFORMATION REQUIRED WITH ENQUIRY AND ORDER

22.1 The information to be supplied by the manufacturer with enquiry and order to the purchaser shall be in accordance with Annex D.

ANNEX A

(*Clause* 2)

LIST OF REFERRED INDIAN STANDARDS

IS No.	*Title*
191: 2007	Copper
335 : 1993	New insulating oils
554 : 1999	Pipe threads where pressure-tight joints are made on the threads — Dimensions, tolerances and designation
1576 : 1992	Solid pressboard for electrical purpose
1608 : 2005	Mechanical testing of metals — Tensile testing
1747 : 1972	Nitrogen
1885 (Part 38) : 1993	Electrotechnical vocabulary: Part 38 Power transformers and reactors
1897 : 2008	Copper strip for electrical purpose
2026	Power transformers :
(Part 1) : 2011	General
(Part 2) : 2010	Temperature rise
(Part 3) : 2009	Insulation levels, dielectric tests and external clearances in air
(Part 5) : 2011	Ability to withstand short circuit
(Part 8) : 2009	Application guide
(Part 10) : 2009	Determination of sound levels
2099 : 1986	Bushings for alternative voltages above 1 000 volts
3024 : 2006	Grain oriented electrical steel sheets and strips
3347	Dimensions for porcelain transformer bushings for use in lightly polluted atmospheres
(Part 1/Sec 1) : 1979	Up to and including 1 kV, Section 1 Porcelain parts
(Part 1/Sec 2) : 1979	Up to and including 1 kV, Section 2 Metal parts
(Part 2/Sec 1) : 1979	3.6 kV Bushings, Section 1 Porcelain parts
(Part 2/Sec 2) : 1979	3.6 kV Bushings, Section 2 Metal parts
(Part 3/Sec 1) : 1988	17.5 kV Bushings, Section 1 Porcelain parts
(Part 3/Sec 2) : 1988	17.5 kV Bushings, Section 2 Metal parts
(Part 4/Sec 1) : 1988	24 kV Bushings, Section1 Porcelain parts
(Part 4/Sec 2) : 1982	24 kV Bushings, Section 2 Metal parts
(Part 5/Sec 1) : 1979	36 kV Bushings, Section 1 Porcelain parts
(Part 5/Sec 2) : 1979	36 kV Bushings, Section 2 Metal parts
3639 : 1966	Fittings and accessories for power transformers

IS No.	*Title*
4253 (Part 2) : 2008	Cork composition sheet: Part 2 Cork and rubber
6162	Paper-covered aluminum conductors:
(Part 1) : 1971	Round conductors
(Part 2) : 1971	Rectangular conductors
7404 (Part1) : 1991	Paper covered copper conductors: Part 1 Round conductors
7421 : 1988	Porcelain bushings for alternating voltages up to and including 1 000 V
8999 : 2003	Pipe threads where pressure tight joints are made on the threads — Verification by means of limit gauges
9335(Part 1) : 1979	Cellulosic papers for electrical purposes: Definitions and general requirements
(Part 2) : 1998	Methods of test
(Part 3/Sec 1) : 1984	Specifications for individual materials, Section 1 General purposes electrical paper
(Part 3/ Sec 3) : 1984	Specifications for individual materials, Section 3 Crepe paper
(Part 3/ Sec 5) : 1985	Specifications for individual materials, Section 5 Special papers
11149 : 1984	Specification for rubber gaskets
12444 : 1988	Continuously cast and rolled electrolytic copper wire rods for electrical Conductors
13730	Specification for particular types of winding wires:
(Part 0/ Sec 3) : 2012	General requirements, Section 3 Enameled round copper wire
(Part 17) : 1996	Polyvinyl acetal enameled rectangular copper wire, Class 105
(Part 27) : 1996	Paper covered rectangular copper wire
16081 : 2013	Insulating liquids — Specification for unused synthetic organic esters for electrical purposes

ANNEX B

(*Clause* 18)

METHOD OF DECLARING EFFICIENCY

B-1 EFFICIENCY

B-1.1 The efficiency to be declared is the ratio of the output in kW to the input in kW and calculated as under.

$$\text{Efficiency } (\eta) = \frac{\text{Output}}{\text{Input}} = \frac{\text{Input} - \text{Total Losses}}{\text{Input}}$$

Total losses comprise:

a) No-load loss, which is considered to be constant at all loads; and

b) Load loss, which varies with load.

The total loss, on load is the sum of (a) and (b).

ANNEX C

(*Clause* 18)

CALCULATION OF INHERENT VOLTAGE REGULATION

C-1 INHERENT VOLTAGE REGULATION

C-1.1 The inherent voltage regulation from no-load to a load of any assumed value and power factor may be computed from the impedance voltage and corresponding load loss measured with rated current in the winding [*see also* IS 2026 (Part 8)].

Let

I = rated current in winding excited;

E = rated voltage of winding excited;

I_{sc} = current measured in winding excited

E_{zsc} = voltage measured across winding excited (impedance voltage);

P_{sc} = watts measured across winding excited

E_{xsc} = reactance voltage = $\sqrt{E_{ZSC}^2 - \left(\frac{P_{SC}}{I_{SC}}\right)^2}$

P = P_{SC} corrected to 75°C, and from current I_{SC} to I;

E_x = $E_{xsc} \times \frac{I}{I_{SC}}$

E_r = $\frac{P}{I}$

$E_x\,\%$ = $100\,E_x/E$;

$E_r\,\%$ = $100\,E_r/E$

$n = I_a/I$; and

I_a = current in the winding excited during the short circuit tests corressponding to that obtained when laoding at the assumed load on the output side and with rated voltage on the input side.

C-1.2 For rated load at unity power factor, the percentage regulation is approximately equal to

$$E_r\,\% + \frac{(E_x\,\%)^2}{200}$$

C-1.3 For rated load any power factor cos ϕ, the percentage regulation is approximately equal to:

$$E_r\,\%\cos\phi + E_x\,\%\sin\phi + \frac{(E_x\,\%\cos\phi - E_r\,\%\sin\phi)^2}{200}$$

C-1.4 For any assumed load other than rated load and unity power factor, the percentage regulation is approximately equal to:

$$n.E_r\,\% + \frac{(n.E_x\,\%)^2}{200}$$

C-1.5 For any assumed load other than rated load and at any power factor cos ϕ, the percentage regulation is approximately equal to:

$$n.E_r\,\%\cos\phi + n.E_x\,\%\sin\phi + \frac{(n.E_x\,\%\cos\phi - n.E_r\,\%\sin\phi)^2}{200}$$

C-1.6 The above formulae are sufficiently accurate for transformers covered by this standard.

ANNEX D

(*Clause* 22.1)

INFORMATION REQUIRED WITH ENQUIRY AND ORDER

A) Normal Information

The following information should be given in all cases:

a) Particulars of the specification to be complied with;
b) Application of Transformer for example normal Distribution Transformer, Solar duty, wind application, Motor starting etc;
c) Single or three phase unit;
d) Number of phases in system;
e) Frequency;
f) Indoor or outdoor type;
g) Type of cooling;
h) Rated power (in kVA);
j) Rated voltages (for each winding);
k) State if tappings are required and if on-load or off-circuit tap-changers, or links are required;
m) Highest voltage for equipment (for each winding);
n) Method of system earthing (for each winding);
p) Insulation level (for each winding), power frequency test level/impulse level;
q) Connection symbol;
r) Neutral terminals, if required (for each winding) and their insulation level to earth;
s) Special requirements of installation, assembly, transport and handling; and
t) Fittings required and an indication of the side from which meters, rating plates, oil-level indicator, etc may be readable.

B) Special Information

The following additional information may be required to be given:

a) If a lightning impulse voltage test is required, whether or not the test is to include chopped waves [*see* IS 2026 (Part 3)];
b) Impedance voltage at rated current, if specific value is required;
c) Altitude above mean sea-level, if in excess of 1 000 m;
d) Whether transformers will be subjected to frequent overcurrent, for example, furnace transformers and traction feeding transformers;
e) Any other exceptional service conditions;
f) Whether noise level measurement is to be carried out;
g) Vacuum withstand of the transformer tank, if a specific value is required;
h) Type of tap-changer controls required (if OLTC is provided);
j) Type of mounting for example pole mounted, ground mounted etc; and
k) Any other appropriate information, including reference to any special tests not referred to above which may be required.

(*Continued from second cover*)

experience is available, separate standard to Distribution transformers filled with Natural/Synthetic Esters shall be brought out.

Considerable assistance has also been taken, while preparing this standard, from IEC 60076-1: 'Power Transformers — Part 1: General'.

Following deviations with respect to IEC standard have been made:

a) Separate standard is prepared for distribution transformers up to rating 2 500 kVA,
b) Ambient conditions are different, and
c) Temperature rises of oil and windings are different.

This standard is intended to cover the generic technical specifications and it does not include all the necessary provisions of a contract. However, the standard calls for agreement between the purchaser and the manufacturer in certain cases.

Unlike, standard on Power Transformers (IS 2026), this standard also purports to recommend standard manufacturing configurations and practices with a view to have standard, reliable and energy efficient distribution transformers in the country.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Bureau of Indian Standards

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Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards : Monthly Additions'.

This Indian Standard has been developed from Doc No.: ET 16 (6686).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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